



Modelling Financial Markets

Chapter 17: Bubbles and Crashes



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EM: Expectation-Maximisation (algorithm)
- ▣ FRED: Federal Reserve Economic Data
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPL: Log-Periodic Power Law
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MC: Monte Carlo (simulation)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States

Chapter Route and Outcomes

Route of this chapter (2×80 minutes)

- ▣ Question: prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value; can statistics detect such episodes **in real time**, and say **when** they end?
- ▣ Session 1: what a bubble is, run-ups and crashes in the data, tests for explosive prices and their date-stamping
- ▣ Session 2: the LPPLS model, regime switching, crypto and AI stocks, a case study, one pass of the AI-assisted research loop
- ▣ Technical detail sits in the Appendix at the end

After this chapter you can

- ▣ separate the fundamental value from a rational bubble and state what can be tested
- ▣ test for explosive prices with GSADF and date-stamp episodes with BSADF, with valid critical values
- ▣ fit LPPLS, judge the stability of its critical time and evaluate its alarms honestly
- ▣ describe booms and busts with a Markov-switching model and test the number of regimes

Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain;
[Wikimedia Commons](#)

- **Tulips, 1636–1637:** contracts on rare bulbs traded at very high prices until the market stopped in February 1637; part of the path fits the fundamentals of a new luxury good [Garber \(1990\)](#)
 - ▶ Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720:** shares rose several-fold within months, then collapsed; a financial innovation, easy credit and a story about future profits [Kindleberger & Aliber \(2005\)](#)
 - ▶ Lesson 2: bubbles come with credit and with high turnover [Scheinkman & Xiong \(2003\)](#)
- **1929:** a boom partly financed with borrowed money ended in the October crash; Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic [Kindleberger & Aliber \(2005\)](#)
 - ▶ Lesson 3: we need tests that work on a single time series, in real time, before the panic

Data Used in This Chapter

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2026	price-dividend ratio
48 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; [Kenneth French Data Library](https://kennethfrench.com) for industries

Fundamental Value

- No-arbitrage alone gives $P_t = \mathbb{E}_t[M_{t+1}(P_{t+1} + D_{t+1})]$, with M the stochastic discount factor and D the dividend
 - ▶ **Model assumption:** a constant required (expected) return r , so $P_t = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{1 + r}$; $\mathbb{E}_t$: expectation under the physical measure, given the information at t
- Solving forward gives the **fundamental value**: $F_t = \sum_{i \geq 1} \frac{\mathbb{E}_t[D_{t+i}]}{(1 + r)^i}$
 - ▶ Gordon case, dividends growing at $g < r$: $F_t = D_{t+1}/(r - g)$
- But $P_t = F_t + B_t$ also solves the equation whenever $\mathbb{E}_t[B_{t+1}] = (1 + r) B_t$
- B_t is the **rational bubble**: investors pay for it because they expect to resell it at a higher price [Blanchard & Watson \(1982\)](#)

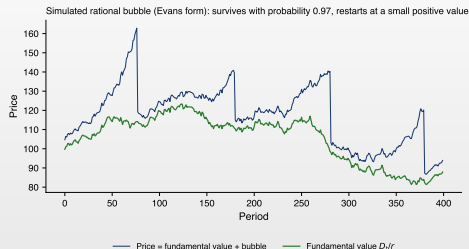
Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$, an **explosive** path
 - ▶ The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- A negative bubble would eventually push the expected price below zero, and a bubble that is zero today must stay zero: rational bubbles are positive and exist from the first trade [Diba & Grossman \(1988\)](#)
- A bubble is the failure of the **transversality condition** $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$: the price contains a value never paid out as dividends (derivation in the Appendix)
- **Testable implication**: if dividends are at most $I(1)$, then without a bubble the price and the price-dividend ratio are **not explosive**
 - ▶ $I(1)$: integrated of order one; the first difference is $I(0)$ (stationary), while the level is not

Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1 - \pi$ [Blanchard & Watson \(1982\)](#)
 - ▶ $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π , $B_{t+1} = \varepsilon_{t+1}$ otherwise
- Expected growth is still r ; while it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time
- After a collapse $B = \varepsilon$ can be negative, which Diba–Grossman rule out: the simulation on the next slide restarts from a positive value (Evans form) [Evans \(1991\)](#)

A Simulated Rational Bubble



- Fundamental value D_t/r , random-walk dividends; $r = 1\%$, $\pi = 0.97$: growth close to 4.1% per period while alive and large, expected life 33.3 periods
- Evans form: collapse to $b_0 = 5$, survival $B_{t+1} = b_0 + \frac{1+r}{\pi}(B_t - \frac{b_0}{1+r})$ times noise of mean 1, so $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$ exactly
- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 1.43 times the fundamental value
- Between episodes the price tracks fundamentals, so a full-sample test sees mostly “normal” data: the tests of this chapter use **windows**

Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017);

CC BY 2.0; Wikimedia Commons

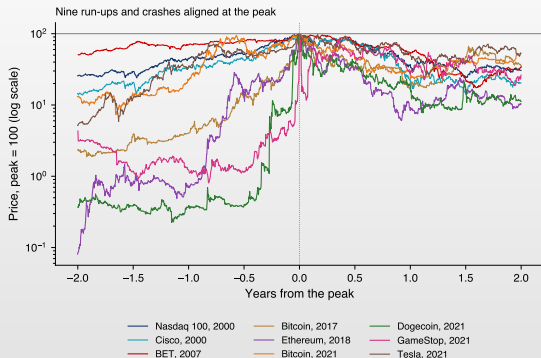
- **Feedback and narratives:** rising prices attract new buyers, “new era” stories spread by word of mouth [Shiller \(2015\)](#)
- **Disagreement and short-sale limits:** the price includes the option to resell to a more optimistic buyer [Scheinkman & Xiong \(2003\)](#)
- **Riding the bubble:** informed traders stay in as long as others do, since no one can burst it alone [Abreu & Brunnermeier \(2003\)](#), [Brunnermeier & Nagel \(2004\)](#)
- All three give prices that grow faster than fundamentals for a while: the explosive signature we will test

Nine Episodes: Run-Up and Crash

Episode	Peak	2-year run-up	Max. fall	Trough	Back at peak
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode; run-up: price change over the two years before the peak; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak



- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ enormously, from +97% (BET) to +124004% (Ethereum); the falls are much more alike, 74–94%
- Common shape: an accelerating rise (convex on the log scale, the growth rate itself rises), a sharp top, a deep fall: the **explosive** behaviour the tests look for
- Selection bias: we picked these episodes because they crashed; many run-ups did not

The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia

[Commons](#)

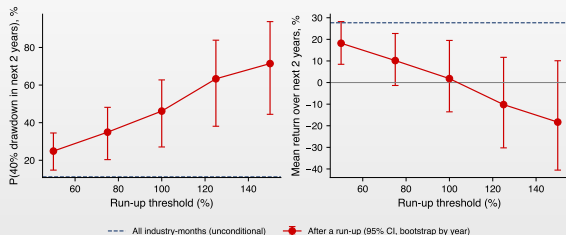
- After the peak of 27 Mar 2000: Nasdaq 100 -83% , Cisco -89% (table); Cisco regained its peak only in Aug 2021
- Hedge funds rode the bubble and cut positions before the fall [Brunnermeier & Nagel \(2004\)](#)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble [Pástor & Veronesi \(2009\)](#)

Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news [Fama \(2014\)](#)
- Greenwood, Shleifer and You answer with **all** US industries since 1926, not with famous episodes only: do sharp run-ups predict crashes, or low returns? [Greenwood, Shleifer & You \(2019\)](#)
- **Run-up**: value-weighted industry return over the past 2 years above a threshold (e.g. 100%), both raw and net of the market
 - ▶ And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted (Section 2)
 - ▶ First month of the run-up; a new episode in the same industry only two years later
- **Crash**: a drawdown of at least 40% from a previous high within the next 2 years
- Our replication, same sample rules: the first 48 of the 49 industries of the [Kenneth French Data Library](#) (without "Other"), months with at least ten firms: 41 679 industry-months, 1926–2026

What Do Sharp Run-Ups Predict?

Bubbles for Fama, 48 US industries 1926-2026: two-year run-ups above the threshold, raw and net of the market

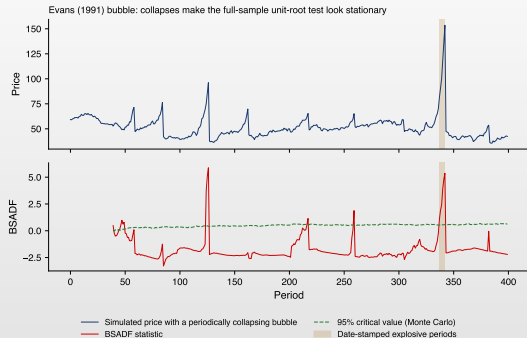


- After a run-up above 100% (52 episodes in 27 industries), a 40% crash followed in 46% of cases, 95% CI [27%, 63%], against 11% for a random industry-month; above 150%: 71% ($n = 21$)
 - CI: bootstrap over calendar years, because run-ups cluster in time (e.g. 1928–1929, 1967–1968, 1980); effective sample size in the Appendix
- The mean return over the next 2 years is not significantly different from zero: 2% after a 100% run-up, CI [−14%, 20%]: no statistically established negative mean return
- Run-ups raise the **risk** of a crash, not the predictable loss: price run-ups alone do not give a profitable timing signal Greenwood, Shleifer & You (2019)

Right-Tailed ADF and Collapsing Bubbles

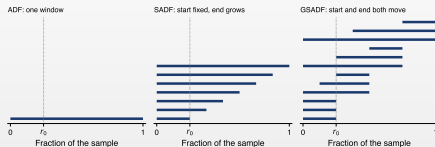
- Regression on the log price y_t : $\Delta y_t = a + b y_{t-1} + \varepsilon_t$; $ADF = \hat{b}/SE(\hat{b})$; here with a constant and no lagged Δy
- Bubble test: $H_0 : b = 0$ (unit root) vs $H_1 : b > 0$ (explosive root), the **right tail**; the left-tail Dickey–Fuller table does not apply [Phillips, Wu & Yu \(2011\)](#)
- Evans: a bubble that collapses periodically makes the **full-sample** series look stationary or like a random walk: low power; collapses give the full-sample slope a negative sign (Appendix) [Evans \(1991\)](#)
- Solution: run the ADF regression on many **windows** and take the largest statistic
 - ▶ SADF (Phillips, Wu and Yu): windows start at the first observation and grow [Phillips, Wu & Yu \(2011\)](#)
 - ▶ GSADF (Phillips, Shi and Yu): windows may start and end anywhere [Phillips, Shi & Yu \(2015a\)](#)

Evans Bubble: Full Sample vs Windows



- 400 simulated periods; BSADF (defined below) against its 95% critical value from 2000 random-walk paths
- Full-sample right-tailed ADF: -5.49 vs 95% critical value -0.09 : no rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46) and GSADF 5.88 (critical value 2.21)
- BSADF crosses its critical value in several bursts, but only one lasts long enough to be date-stamped: the minimum-duration rule trades short bursts for fewer false alarms

Three Ways to Choose Windows: SADP, GSADF, BSADF

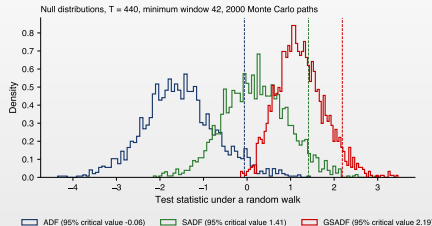


- $ADF'_{r_1}^2$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample; each bar in the chart is one ADF regression
- $SADF(r_0) = \sup_{r_2 \in [r_0, 1]} ADF_0'^2$ Phillips, Wu & Yu (2011)
- $GSADF(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}'^2$ Phillips, Shi & Yu (2015a)
- Backward SADF at the end point r_2 : $BSADF_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}'^2$; $GSADF = \sup_{r_2} BSADF_{r_2}$
 - $BSADF_{r_2}$ uses only data up to r_2 : it can be computed **in real time**
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$, e.g. $T = 440$ months: $r_0 = 0.096$, 42 months Phillips, Shi & Yu (2015a)

Date-Stamping: When Did It Start and End?

- **Origination:** the first date at which BSADF exceeds its critical value Phillips, Shi & Yu (2015a)
 - ▶ The critical value sequence cv_{r_2} : 95% quantile of $BSADF_{r_2}$ under the null, date by date
- **Termination** (PSY, eq. 8): the first date after $\hat{r}_e + \delta \ln(T)/T$ at which BSADF falls below its critical value, δ depending on the frequency Phillips, Shi & Yu (2015a)
- **Rule used in this chapter:** the minimum-duration filter of `exuber` (`datestamp`, `min_duration`), which drops runs of exceedances shorter than L Vasilopoulos, Pavlidis & Martínez-García (2022)
 - ▶ $L = \lceil \ln T \rceil$ ($\delta = 1$); $T = 440$: 7 months. Start: first exceedance; end: last exceedance (`exuber` reports the next date). Eq. 8 would instead extend a one-date blip to L dates
 - ▶ Real time: the start is dated in hindsight; the episode is **confirmed** only after L consecutive exceedances, $L - 1$ periods later
- PWY date-stamping uses the recursive ADF_0^2 instead: it may miss a second bubble after a collapse Phillips, Shi & Yu (2015b)

Null Distributions and Critical Values



- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2000 paths of $T = 440$; 95% critical values: a single ADF -0.06 (negative!), SADF 1.41, GSADF 2.19 [Phillips, Shi & Yu \(2015a\)](#)
- A supremum over more windows shifts the distribution to the right: the GSADF bar is the price of searching over about 80 000 windows; simulate it for each T , r_0 and lag order
- Asymptotics (Appendix): the null limit is pivotal; against mildly explosive roots the tests are consistent; date-stamping keeps a 5% false-alarm chance at every date [Phillips, Shi & Yu \(2015b\)](#)

Changing Volatility: Wild Bootstrap

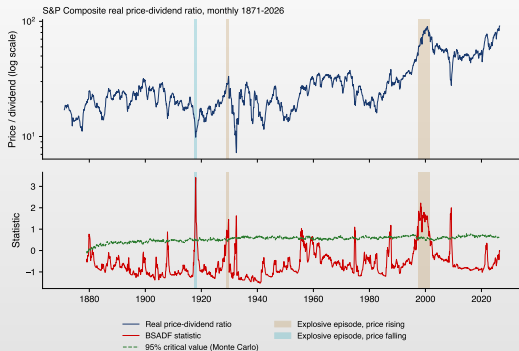
- ▣ Monte Carlo critical values assume shocks with constant variance; real prices have volatility clusters and shifts
- ▣ A burst of volatility can mimic explosiveness and inflate the rejection rate [Harvey et al. \(2016\)](#)
- ▣ **Wild bootstrap**: multiply the observed, demeaned changes Δy_t by random signs $w_t = \pm 1$, rebuild the path, recompute BSADF [Phillips & Shi \(2020\)](#)
 - ▶ The bootstrap null keeps the volatility pattern of the data but has no explosive root
 - ▶ Centred with the full-sample mean and resampling the whole path: **retrospective** critical values; in real time, recompute them with data up to each date
- ▣ Validity: the bootstrap data have a unit root by construction and inherit the time pattern of $\text{Var}(\Delta y_t)$; demeaning removes the drift, as in the null model [Harvey et al. \(2016\)](#)

Size of GSADF When Volatility Changes: Monte Carlo

Null model, no bubble	GSADF, MC cv	GSADF, wild cv	False episode, MC	False episode, wild
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%
Volatility $\times 1/3$ at $T/2$	4.3%	6.1%	48.0%	54.8%
GARCH(1,1), $\alpha = 0.10$, $\beta = 0.88$	16.6%	6.8%	58.1%	52.8%

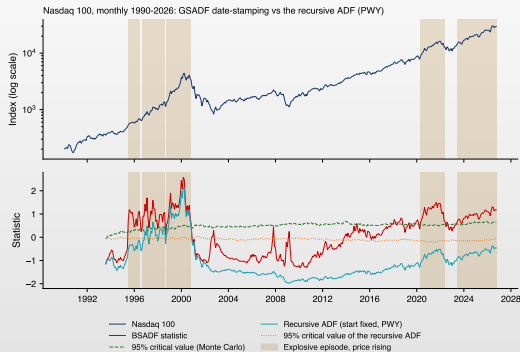
- Rejection rates at 5%; $y_t = T^{-1} + y_{t-1} + e_t$, $T = 400$, $w_0 = 40$, 1 000 paths per row, each with its own wild bootstrap (199 replications); Monte Carlo error ± 1.4 points near 5%; false episode: BSADF above its pointwise 95% value for 6 periods
- A late rise in volatility is mistaken for explosiveness (32.0% with Monte Carlo, 4.1% with wild critical values); under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless [Harvey et al. \(2016\)](#)
- Even with constant volatility 49.1% of the paths contain a false date-stamped episode: the size is controlled date by date, not over the sample

S&P 500 Price-Dividend Ratio, 1871–2026



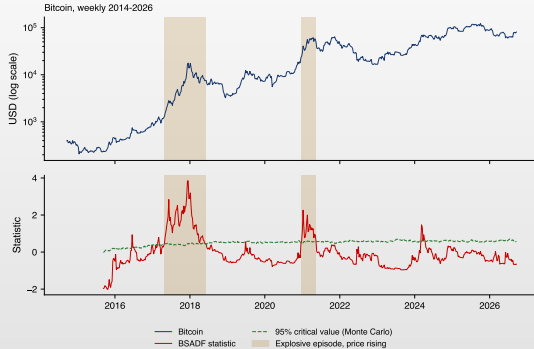
- Real price over real dividend, monthly, Jan 1871–Jun 2026, $T = 1865$, minimum window 96 months
- GSADF = 3.41 vs 95% critical value 2.39 (99%: 2.88); the full-sample ADF (-1.11) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
 - ▶ 1917–1918 is an accelerating **fall** of the ratio (real prices fell faster than real dividends): explosive, but not a bubble
- Dot-com: peak 90.2 in Aug 2000 vs a long-run mean of 29.8; the sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not an explosive one
- A ratio can explode through its denominator: always check both price and dividend

Nasdaq 100: GSADF vs PWY Date-Stamping



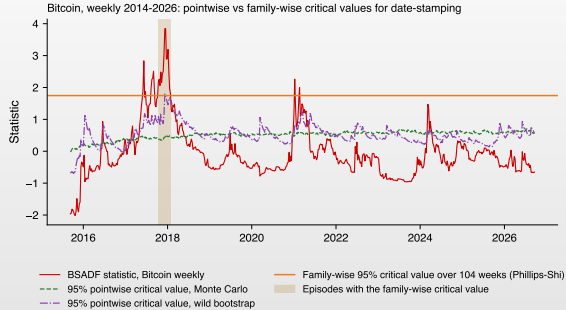
- Monthly log closes 1990–2026, $T = 440$, minimum window 42 months; 2 000 Monte Carlo paths
- GSADF = 2.57 (95% critical value 2.19); SADF = 2.14 (critical value 1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
 - PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF starts in 1990 and is dragged down by the long flat period after 2000: it misses the post-2020 episodes that BSADF finds
- Dot-com episode dated from Jun 1995, 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995

Bitcoin: Explosive Episodes



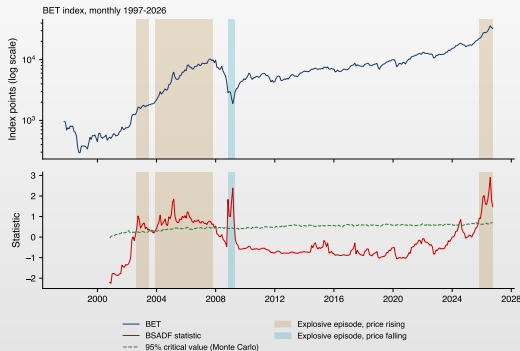
- Weekly log closes 2014–2026, $T = 626$, minimum window 51 weeks; Monte Carlo critical values
- GSADF = 3.85: Monte Carlo 95% critical value 2.27, wild bootstrap 2.82: rejected either way
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: episode dated from Apr 2017, 8 months before the peak of 16 Dec 2017, confirmed in real time in Jun 2017; it ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged: steady growth is not the same as explosive acceleration

Date-Stamping Is a Multiple Test: Bitcoin



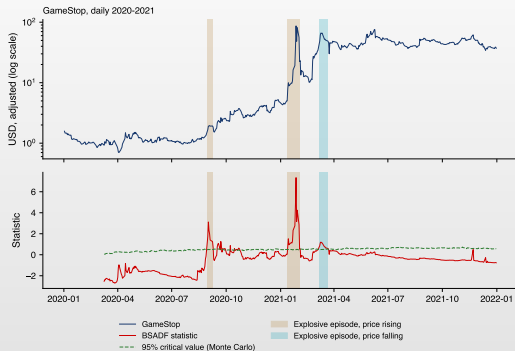
- BSADF is compared with its 95% critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER): the probability of at least one false alarm over the whole monitoring window; in 2 000 random walks with the Bitcoin T and w_0 : a crossing in 95%, an “episode” in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF over τ_b dates; family-wise critical value 1.75, against pointwise values of at most 0.71 [Phillips & Shi \(2020\)](#)
- Episodes above the higher threshold: Oct 2017–Jan 2018; the 2020–2021 episode found with pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the Appendix)

BET: The 2007 Boom and 2025



- Monthly log closes of the official BET, 1997–2026, $T = 348$, minimum window 37 months
- $GSADF = 2.90$ vs 95% critical value 2.21; episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up before and around Romania's accession to the European Union (January 2007); BET then fell 83% from its peak of 24 Jul 2007
- Oct 2008–Mar 2009 is an accelerating **fall**: explosive can mean down as well as up
- The current episode (from Oct 2025) is still open: explosive does not mean it will crash; it means the rise is accelerating

GameStop, January 2021



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- Daily log closes 2020–2021, $T = 504$, window 45 days; GSADF = 7.34, critical value 2.21
- January episode dated from 13 Jan 2021, 14 days before the closing high of 27 Jan 2021; confirmed on 22 Jan 2021
- A surge of weeks: it needs daily data; the test does not identify the mechanism (short covering)

 MFM_ch17_bet_gamestop

Real Time vs Hindsight

Episode	Start / Peak	Confirmed	Lead (m.)	Lead, conf.	Rise after conf.	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	Dec 1995	57	51	+716%	−76%
BET, monthly	Nov 2003 / Jul 2007	Apr 2004	44	39	+299%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	Jun 2017	8	6	+590%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	Feb 2021	4	2	+66%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	22 Jan 2021	0.5	0.2	+435%	−88%

- Start: first exceedance of the chain of BSADF episodes that leads to the peak (hindsight); confirmed: after L consecutive exceedances, the first date known in real time; leads in months; rise: confirmation to peak; fall: to the lowest close within 2 years (definitions in the Appendix)
- Even at confirmation the tests **warn early**, but the lead ranges from 5 days to about 4.2 years, and selling at confirmation would have missed most of the rise: they do not time the crash [Phillips, Shi & Yu \(2015b\)](#)

Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0
(Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS): imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches [Johansen, Ledoit & Sornette \(2000\)](#)
- Rational investors stay in only if the price grows fast enough to pay for the rising crash risk
- Result: **faster-than-exponential** growth, decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date [Sornette \(2017\)](#)

The LPPLS Equation

- $\ln p(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
- A : log price at t_c ; $B < 0$: the super-exponential trend for a rising bubble; $0 < m < 1$: power-law exponent
 - ▶ The growth rate of the power-law trend, $-Bm(t_c - t)^{m-1}$, rises without bound as $t \rightarrow t_c$
- C, ω, ϕ : amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at t_c

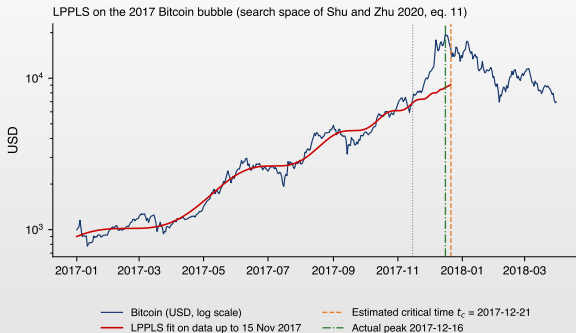
Calibration: Filimonov and Sornette

- Rewrite the cosine: $C \cos(\omega \ln(t_c - t) - \phi) = C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))$ Filimonov & Sornette (2013)
 - ▶ A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS
- Search only over (t_c, m, ω) : a grid, then a Nelder–Mead refinement of the sum of squared residuals
- Search space of Shu and Zhu (Section 2.3, eq. 11): $m \in [0, 1]$, $\omega \in [1, 50]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$, damping $\frac{m|B|}{\omega|C|} \geq 1$, an approximate non-negativity condition for the crash hazard rate, the crash probability per unit time given no crash yet (exact: $m|B| \geq |C|\sqrt{m^2 + \omega^2}$, Seminar 17, A8) Shu & Zhu (2020)
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Qualified Fits: The Filter

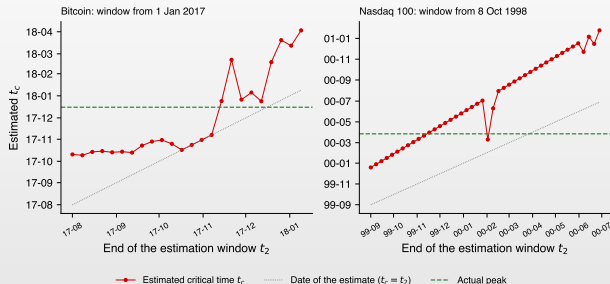
- A fit on $[t_1, t_2]$ is **qualified** if it passes all conditions of Shu and Zhu (Section 2.3, eq. 12), and $B < 0$ for a rising bubble Shu & Zhu (2020)
 - ▶ $m \in [0.01, 0.99]$, $\omega \in [2, 25]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
 - ▶ Number of oscillations (half-periods) $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not noise
 - ▶ Largest relative error of the fitted price $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
 - ▶ Lomb periodogram of the detrended residual $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$: log-periodicity significant at 10%
 - ▶ Residual unit-root check for $\ln \hat{p}_t - \ln p_t$: Dickey–Fuller and Phillips–Perron reject a unit root at 10% (a stationary, mean-reverting residual; this alone does not make it AR(1))
- $[t_1, t_2]$: the estimation window; the filter is applied to each window separately

LPPLS on Bitcoin, 2017



- Fit on daily log prices from 1 January to 15 November 2017; the path after 15 November is out of sample
- The fit follows the 2017 rise and its oscillations closely: $m = 0.95$, $\omega = 10.54$, $B = -2.62$
- Estimated t_c : 21 Dec 2017; actual peak: 16 Dec 2017, i.e. 5 days earlier
- The fit fails the condition on relative error ≤ 0.15 : it does **not** pass the filter, even though it looks convincing

How Stable Is the Critical Time?

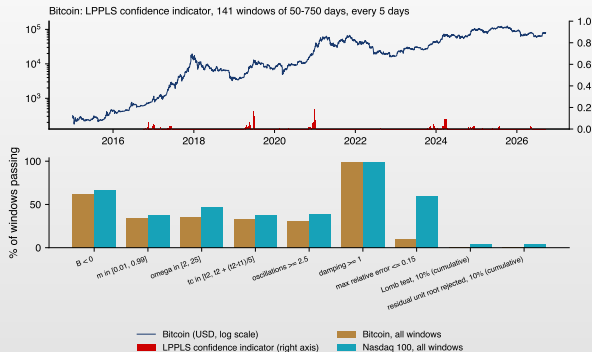


- A good in-sample fit is not a forecast: each point is one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits) t_c ranged from Oct 2017 to Feb 2018; mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Dec 1999 to Sep 2000; mean absolute error 70 days; $t_c \geq t_2$ is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin) and 0% (Nasdaq 100); the birth of a bubble is easier to diagnose than its burst [Demos & Sornette \(2017\)](#)

The LPPLS Confidence Indicator

- At each date t_2 , fit many windows $[t_2 - L, t_2]$ and report the **share of qualified fits** [Demirer et al. \(2019\)](#)
 - ▶ Grid of Shu and Zhu (Section 2.3): 141 windows, L from 750 down to 50 observations in steps of 5; t_2 every 5 observations; data up to t_2 only [Shu & Zhu \(2020\)](#)
 - ▶ Daily closes; Bitcoin trades every day, so one observation is one calendar day
- A high share means the bubble signature is robust to the choice of window
- Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
 - ▶ Fall measured from the price on the signal date; only dates whose 90 calendar days are fully observed (the last three months of data are dropped)

Bitcoin: Real-Time Confidence Indicator



- Top: share of the 141 windows with a qualified fit, every 5 days, 2016–2026; bottom: share of windows passing each condition (Lomb and residual tests cumulative)
- Bitcoin: the indicator was positive on 14% of the 2017 dates, 3% in 2021 and 12% in other years; maximum 0.18 in Dec 2020
- After a signal, a 20% fall within 90 days followed in 22% of cases, vs 37% without a signal (709 dates)
 - Nasdaq 100, 1997–2026: 4% after a signal vs 11% without (1482 dates)
- The raw difference is not yet evidence: signals come in runs and the 90-day outcome windows overlap (test on the next slide)

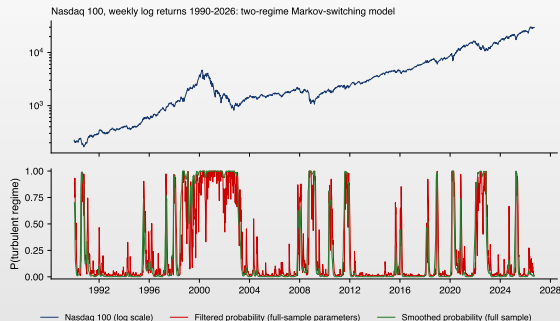
Is the Indicator Better than Chance? A Test

- $H_0: P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$; statistic $\Delta = \hat{\rho}_{\text{on}} - \hat{\rho}_{\text{off}}$
- Dates 5 days apart share up to 85 of their 90 outcome days: the two-sample z-test, which assumes independent dates, understates the standard error
- Circular block bootstrap of the (signal, outcome) pairs, blocks of 90 days, 9 999 replications; linear probability regression with Newey–West standard errors
 - ▶ Bitcoin: $\Delta = -15.5$ points, 95% CI $[-30.0, 1.8]$; block SE 8.1 vs independent-data SE 5.0; regression $p = 0.049$
 - ▶ The same test for the Nasdaq 100, with a power calculation: Seminar 17, Exercise B6
- The interval does not lie above zero: in our data a signal is **not** followed by a large fall more often than no signal; the point estimate is negative (regression $p = 0.049$): signals come in steady rises, not just before crashes

The Markov-Switching Model

- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t}\varepsilon_t$, $\varepsilon_t \sim N(0, 1)$, regime $s_t \in \{1, 2\}$ unobserved [Hamilton \(1989\)](#)
 - ▶ Each regime has its own mean μ and volatility σ : a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$; expected duration of regime i : $1/(1 - p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates $P(s_t = j \mid r_1, \dots, r_t)$ week by week
- **Filtered** probability: returns up to t for given parameters; **smoothed**: the whole sample (hindsight)
 - ▶ Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to t
 - ▶ A worked filter step is in Seminar 17, Exercise A5

Nasdaq 100: Two Regimes

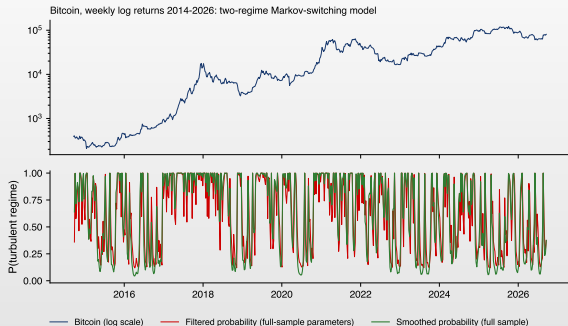


Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{\rho}_{ii}$	Dur.
Calm	0.43 (0.06)	2.30	0.987	75.4
Turbulent	-0.32 (0.27)	5.40	0.957	23.5

- Weekly log returns in %, 1990–2026, 1 915 weeks; duration in weeks; calm regime 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5); filtered probability 0.99 at the end of March 2000: prices kept rising while volatility jumped

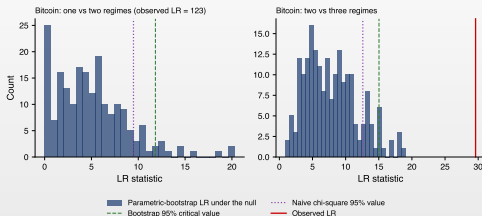
 MFM_ch17_markov_switching

Bitcoin: Two Regimes



- Weekly log returns 2014–2026, 626 weeks
- Calm: $\hat{\mu} = 0.26\%$ per week, $\hat{\sigma} = 2.93\%$, duration 3.8 weeks
 - ▶ Turbulent: $\hat{\mu} = 1.20\%$ (SE 0.58), $\hat{\sigma} = 11.22\%$, duration 6.3 weeks, 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean; regimes are short: volatility states, not bubble phases
- Regimes ignore the price level (a slow, calm bubble stays calm): BSADF says “the price is accelerating”, the regime says “risk has changed”

How Many Regimes? A Bootstrap LR Test

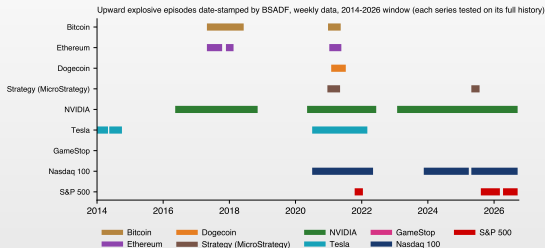


- H_0 : one regime (iid Normal returns) vs H_1 : two Markov regimes, $LR = 2(\hat{\ell}_2 - \hat{\ell}_1)$; not χ^2 : p_{11}, p_{22} are not identified under H_0 (Appendix Hansen (1992))
- **Parametric bootstrap**: simulate from the fitted H_0 model, refit both models, recompute LR; 199 replications;
 $p = (1 + \#\{LR^* \geq LR\}) / (B + 1)$
- One vs two: Bitcoin LR = 123.2 (95% value 11.8, $p = 0.005$); Nasdaq 100 LR = 475.7 (11.1, $p = 0.005$)
- Two vs three, Bitcoin: LR = 29.6, bootstrap 95% value 15.0 vs naive χ^2_6 value 12.6; $p = 0.005$ (Seminar 17, B8)
- One regime is rejected for both series; a third Bitcoin regime is also supported. Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes

Are Crypto-Assets Bubbles?

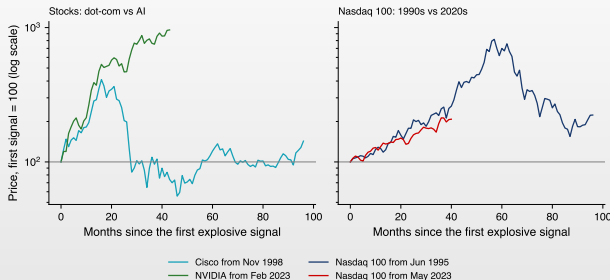
- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero [Cheah & Fry \(2015\)](#)
- Alternative: value from network adoption (Metcalf's law), with LPPL bubbles on top [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble [Pele et al. \(2023\)](#)
- Without a fundamental value we can still test the price: is it explosive, and when?

Explosive Episodes, 2014–2026



- Weekly log closes; each series tested on its own history with its own Monte Carlo critical values; rising episodes only
- Crypto: Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021: booms cluster in 2017 and late 2020–2021
- Strategy (a company that holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
- GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration; an accelerating **fall** also gives $b > 0$: always report the direction

Dot-Com vs AI



- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
- Cisco: first signal Nov 1998, peak after 16 months at 4.1 times the signal price; NVIDIA: first signal Feb 2023, 43 months later at 9.6 times, episode still open
- Nasdaq 100 from Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months, a slower rise than in the 1990s

Is There an AI Bubble? What the Tests Can Say

- ▣ Monthly BSADF flags NVIDIA (Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing) and the Nasdaq 100 (current episode from May 2023)
- ▣ The tests cannot separate a bubble from explosive fundamentals: fast-growing expected profits of a new technology can also make prices explosive [Pástor & Veronesi \(2009\)](#)
- ▣ History: Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- ▣ What we can say: the tests detect explosive price dynamics; the industry run-up evidence of Bubbles for Fama motivates watching crash risk but does not quantify it for these stocks, and the timing is unknown

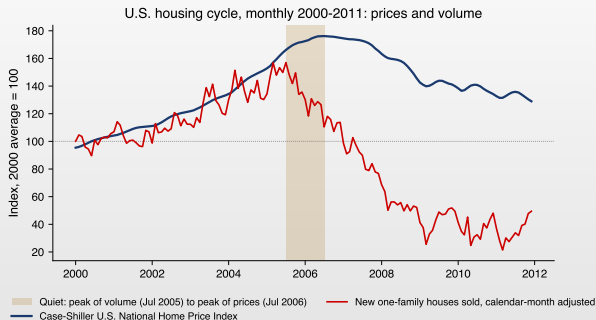
Case Study (1/5): the Research Question

- Paper: [DeFusco, Nathanson & Zwick \(2022\)](#), *Journal of Financial Economics*, 146(1), 205–229
 - ▶ Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Question: who buys during a boom, and can speculation explain the joint path of prices, transaction volume and unsold listings (Introduction)?
 - ▶ Speculation, in the Harrison–Kreps definition the authors adopt (footnote 1): paying more for an asset because one has the right to resell it
- Three phases (Section 2): the boom (prices and volume rise), the “quiet” (volume falls while prices still rise and unsold listings pile up), the bust
- Why it matters: the explosiveness tests of this chapter use prices only; Bubbles for Fama [Greenwood, Shleifer & You \(2019\)](#) also studies turnover, issuance and volatility, and bubbles come with high turnover [Scheinkman & Xiong \(2003\)](#)
 - ▶ Quantities show who drives a boom, as blockchain flows did for the 2017 Bitcoin boom [Griffin & Shams \(2020\)](#)

Case Study (2/5): Data, Method and Headline Results

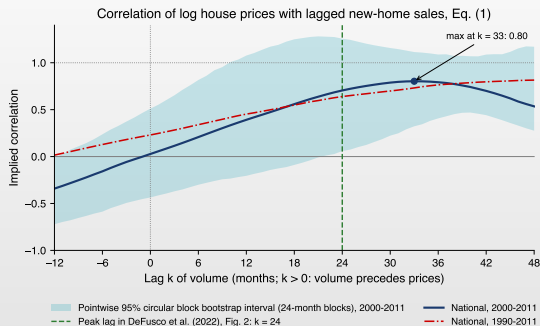
- Data (Section 1.1): 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas; **speculators**: short-term buyers who resell within three years, and non-occupant buyers (Section 1.2)
- Lead-lag (Eq. 1): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$, $k = -12, \dots, 48$ months, p the demeaned log price, v the demeaned sales rate; implied correlation $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$
- Figure 2: the correlation peaks at a lag of 24 months; Figure 4: short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: one standard deviation more speculative volume goes with a price boom larger by 24.9 points and a bust deeper by 7.4 points: cities with more speculation had larger booms and larger busts
- Model (Sections 5–6, Table 6): extrapolating buyers with short horizons amplify the cycle; a 5% purchase tax cuts the bust from 8.2% to 1.1%

Case Study (3/5): Prices and Volume, 2000–2011



- FRED: Case-Shiller U.S. National Home Price Index and new one-family houses sold, 2000 average = 100; volume adjusted by calendar-month means, as in Figure 1 of [DeFusco, Nathanson & Zwick \(2022\)](#)
- Boom: sales peak at 157 in Jul 2005 (paper: 150% of the 2000 level); prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak, when sales are already down to 110 (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices; new-home sales exclude resales, so they cannot show who trades

Case Study (4/5): Volume Leads Prices



- Eq. (1) on the national series, without a constant; pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- 2000–2011: maximum at $k = 33$ (0.80, CI [0.32, 1.11]); at the paper's $k = 24$ ours is 0.70, CI [0.06, 1.26]
- The band excludes zero for $k = 22, \dots, 48$, but one national cycle holds few independent blocks: wide intervals
- Direction as in the paper (volume first, by two to three years); its length is not precise (1990–2011: $k = 48$)

Case Study (5/5): Limits and an Open Question

- Data limit of our illustration: the public series are national and cover new homes only, so short-term and non-occupant volume cannot be built; Figures 1–2 can be illustrated, Table 2 cannot (Seminar 17, C3)
- Limits stated by the authors
 - ▶ Holding periods are realised, not intended (footnote 4; instrumental-variable check in the Online Appendix)
 - ▶ The non-occupant proxy misclassifies owners who give the property address for tax mail (Section 1.2)
 - ▶ The counterfactuals may overstate the role of short-term buyers, because selling horizons are exogenous in the model; credit and new construction are left out (Sections 4.3, 6.5.3, 7)
- Open question: does trading volume also lead prices in crypto booms, and is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

An Open Question

Can bubble alarms be raised in real time with a controlled false-alarm rate, and do they carry crash information?

- BSADF date-stamping runs one right-tailed test per date: the size is controlled date by date, not over the whole monitoring period
 - ▶ Live research: monitoring rules for bubbles *and* crashes with controlled false alarms Whitehouse, Harvey & Leybourne (2023); real-time “econometric thermometers” for housing Shi & Phillips (2023)
- Formal version, for an alarm A_t and a crash $C_{t,h}$ (a fall of at least 20% within $h = 90$ days):
 $H_0: P(C_{t,h} | A_t = 1) = P(C_{t,h} | A_t = 0)$, H_1 : larger after an alarm
 - ▶ Falsified if the hit rate after alarms does not exceed the rate without alarms, under inference that respects overlapping windows
- Identification: explosive prices need not be a bubble
 - ▶ Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009); run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

The Discovery Loop and Its Checks

- **Literature:** ask for papers on the false-alarm rate of sequential bubble monitoring; resolve every DOI on [Crossref](#) and find the table the answer quotes
- **Hypotheses:** rational bubble, explosive expected cash flows, falling discount rate, each with its own observable (e.g. an explosive price-to-earnings ratio vs explosive earnings)
- **Code:** replicate a known number first ($GSADF = 3.85$ for weekly Bitcoin), then extend
- **Robustness:** pre-register the grid of horizons and crash thresholds with a Holm correction; **critique:** ask the assistant to attack the result as a referee
- **Checks** the researcher must run (details and prompts in the Appendix)
 - ▶ No full-sample information in a date stamp; right-tail critical values; wild bootstrap under changing volatility; multiplicity over dates and assets; inference robust to overlapping outcome windows

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal: “a positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher’s exact test”; alarms come in runs and outcomes overlap, so we test by circular rotations
- Bitcoin (709 dates, every 5 days): a fall of 20% within 90 days after 22% of alarms vs 37% without
 - ▶ Difference -15.5 percentage points; one-sided $p = 0.96$ for H_1 from 658 rotations of the alarm series
 - ▶ Nasdaq 100: 4% vs 11%, difference -7.1 points, $p = 0.98$
- Pre-registered grid of 4 horizons $\times$ 3 thresholds: smallest raw p is 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
 - ▶ No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation: the indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal (assumptions in the Appendix)

Your Turn: a Project Seed

- **Question:** can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period and still warn before large falls?
- **Replicate first:** GSADF = 3.85 for weekly Bitcoin, then the crash probability after run-ups above 100% in [Greenwood, Shleifer & You \(2019\)](#) (ours: 46%)
 - ▶ Extend: calibrate the alarm threshold by simulation over the whole monitoring period [Whitehouse, Harvey & Leybourne \(2023\)](#); evaluate the alarms with rotations or a block bootstrap
- **Deliverables:** a pre-registered grid committed before the evaluation, a notebook that regenerates every number, false-alarm rates and hit rates with valid p -values
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it in AI_ERRORS.md ([templates](#))

Key Takeaways

- ▣ A bubble is price minus fundamental value; rational bubbles must be explosive, and bursting bubbles hide from full-sample tests
- ▣ GSADF tests for explosive episodes, BSADF date-stamps them in real time; with a late volatility rise Monte Carlo critical values reject 32.0% of true nulls, the wild bootstrap 4.1%; date-stamping needs family-wise control
- ▣ Across our episodes the tests warned months before the peak, but did not time the crash
- ▣ LPPLS describes the acceleration; its critical time moves with the window, and the confidence indicator is judged by a dependence-robust test of hit rates
- ▣ Sharp run-ups raise crash risk without a statistically established average loss: a risk-management signal, not a trading rule

Which Tool When?

Question	Tool
Has there been any explosive episode? When did it start and end?	GSADF with simulated critical values BSADF date-stamping, minimum duration $\lceil \ln T \rceil$; family-wise critical values for long monitoring
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets (Bubbles for Fama)

Check Yourself, and Next: Chapter 18

Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically?
- Why is the 95% critical value of GSADF far above that of a single ADF?
- Why is a single LPPLS critical time not a crash forecast?
- Why must hit rates after alarms be tested with a block bootstrap or rotations?

Chapter 18: systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- CoVaR and SRISK measure how much one institution adds to the risk of the system
- Further reading: [Franke, Härdle & Hafner \(2019\)](#), [Kindleberger & Aliber \(2005\)](#), [Shiller \(2015\)](#), [Sornette \(2017\)](#)

Appendix: Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself h times: $P_t = \sum_{i=1}^h \frac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \frac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$ (law of iterated expectations)
- Let $h \rightarrow \infty$: $P_t = F_t + B_t$ with $B_t = \lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition:** $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$; it holds $\Leftrightarrow B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period: $\mathbb{E}_t[B_{t+1}] = \lim_h \mathbb{E}_t[P_{t+1+h}]/(1+r)^h = (1+r) B_t$, the martingale condition of the rational bubble
- Seminar 17, Exercise A1 derives the general solution and why $B_t \geq 0$ rules out a bubble that starts from zero [Diba & Grossman \(1988\)](#)

Appendix: Bubbles for Fama: Effective Sample Size

- ▣ The run-ups fall in 31 calendar years and overlap: consecutive 2-year windows of the same boom share months
- ▣ Year-block bootstrap SE (9.1 percentage points) vs binomial SE (6.9): a design effect of 1.7, i.e. about 30 independent observations
- ▣ Blocks of 2, 3 and 5 consecutive years (2-year windows of adjacent years overlap): $SE = 8.9, 8.7, 8.3$
- ▣ Unconditional mean: 28%, above the CI of the mean return after a 100% run-up (−14% to 20%): returns after run-ups look lower than usual, but this is a descriptive comparison, not a formal test of the difference of two dependent estimates

Appendix: Why the Full-Sample ADF Fails: the Leverage of Collapses

- OLS slope: $\hat{b} = \sum_t \tilde{y}_{t-1} \Delta y_t / \sum_t \tilde{y}_{t-1}^2$, with $\tilde{y}_{t-1}$ the demeaned lagged log price
- Expansion: $\Delta y_t > 0$ while $\tilde{y}_{t-1}$ rises above its mean: positive terms
- Collapse: the most negative Δy_t occurs at the highest $\tilde{y}_{t-1}$: a few terms of the largest size and the opposite sign
- Evans form in logs, $y = \ln B$, collapse to $\delta = e^c$ with probability $1 - \pi$; for $B \gg \delta$ the survival log growth is $\approx \ln \frac{1+r}{\pi}$:
 $\mathbb{E}[\Delta y_t | y_{t-1}] \approx \pi \ln \frac{1+r}{\pi} + (1 - \pi)(c - y_{t-1})$, slope $-(1 - \pi) < 0$: apparent mean reversion although $\mathbb{E}_t B_{t+1} = (1 + r)B_t$
 Evans (1991)
- Windows that contain the expansion but not the collapse have $\hat{b} > 0$ when the log price accelerates there (next slide): a supremum over windows restores the power Phillips, Shi & Yu (2015b)

Appendix: Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**: $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$, an autoregressive root $1+r > 1$
 - ▶ In logs a pure bubble path $B_t = B_0(1+r)^t$ has constant growth: $\ln B_t = \ln B_0 + t \ln(1+r)$, a deterministic linear trend in logs, $b = 0$ in $\Delta y_t = a + b y_{t-1} + \varepsilon_t$
- The log price $\ln(F_t + B_t)$ **accelerates** while the bubble share grows: its growth rate rises from that of F towards r (a convex log path)
 - ▶ Once the bubble dominates, log growth is roughly constant again: the log-price regression then has little power
- So the log-price test targets a different alternative, **accelerating** log prices, not the level condition itself; the specification of the null and of the deterministic terms matters [Phillips, Shi & Yu \(2014\)](#)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq); PSY test the price-dividend ratio [Phillips, Wu & Yu \(2011\)](#), [Phillips, Shi & Yu \(2015a\)](#)

Appendix: Limit Theory (1): the Null Distribution

- Null model: $y_t = d T^{-\eta} + y_{t-1} + \varepsilon_t$, $\eta > 1/2$, ε_t iid $(0, \sigma^2)$: a random walk with an asymptotically negligible drift [Phillips, Shi & Yu \(2015a\)](#)
- By the functional central limit theorem and continuous mapping (PSY, Theorem 1, eq. 5), with $r_w = r_2 - r_1$ and W a standard Brownian motion:
 - ▶ $\text{ADF}_{r_1}^2 \Rightarrow \frac{\frac{1}{2} r_w [W(r_2)^2 - W(r_1)^2 - r_w] - \int_{r_1}^{r_2} W dr [W(r_2) - W(r_1)]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W^2 dr - \left(\int_{r_1}^{r_2} W dr \right)^2 \right\}^{1/2}}$
- σ and d disappear: the limit is pivotal. SADF and GSADF are suprema of the same functional, so their critical values depend only on r_0 and the deterministic terms
- $r_1 = 0$, $r_2 = 1$ gives the Dickey–Fuller limit with a constant: its 95% quantile is negative (simulated: -0.06), far from 1.645
- The drift and intercept specification changes the limit and the size [Phillips, Shi & Yu \(2014\)](#)

Appendix: Limit Theory (2): Mildly Explosive Alternatives

- Alternative: $y_t = \rho_T y_{t-1} + \varepsilon_t$ with $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \rightarrow \infty$, $k_T = o(T)$; PSY use $k_T = T^\alpha$, $\alpha \in (0, 1)$
 - ▶ Closer to 1 than any fixed explosive root, farther than a local-to-unity root $1 + c/T$: the “mild” speed of real bubbles
- Phillips and Magdalinos: $\frac{\rho_T^T}{\rho_T^2 - 1} (\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$, a standard Cauchy law, without Gaussian errors [Phillips & Magdalinos \(2007\)](#)
 - ▶ Rate $k_T \rho_T^T$: exponentially fast; the ADF statistic diverges to $+\infty$
- Hence SADF and GSADF diverge too (for GSADF any window of at least $r_0 T$ observations inside the bubble suffices): the tests are consistent, power $\rightarrow 1$ [Phillips, Shi & Yu \(2015b\)](#)
- Finite samples: power depends on c , α and the duration; short bursts may stay below the bar, as in the Evans simulation

Appendix: Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8): $\hat{r}_e = \inf\{r_2 \geq r_0 : \text{BSADF}_{r_2} > \text{scv}_{r_2}^{\beta_T}\}$,
 $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \text{BSADF}_{r_2} < \text{scv}_{r_2}^{\beta_T}\}$
- $\text{scv}_{r_2}^{\beta_T}$: the $100(1 - \beta_T)\%$ quantile of BSADF on $\lfloor Tr_2 \rfloor$ observations
 - Consistency, $(\hat{r}_e, \hat{r}_f) \rightarrow_p (r_e, r_f)$, needs $\frac{1}{\text{scv}^{\beta_T}} + \frac{\text{scv}^{\beta_T}}{T^{1-\alpha/2}} \rightarrow 0$ (PSY, eq. 11)
- The critical value must diverge, so that the false-alarm probability vanishes ($\beta_T \rightarrow 0$), but slower than $T^{1-\alpha/2}$, so that true bubbles are still found; e.g. growing like $\ln T$ [Phillips, Shi & Yu \(2015a\)](#)
- In practice β_T is fixed at 5% at every date: each date keeps a 5% false-alarm chance, and over hundreds of dates a false alarm becomes likely (size study below)

Appendix: The Price-Dividend Ratio: Comparison with PSY

- ▣ Our sample: 1 866 monthly levels, Jan 1871–Jun 2026; critical values from 1 000 Monte Carlo paths
- ▣ PSY, on data to 2010 and without our filter of 8 consecutive months, also date 1879–1880, 1955–1956 and 1986–1987 [Phillips, Shi & Yu \(2015a\)](#)
- ▣ With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too (Seminar 17, B1)

Appendix: Family-Wise Critical Values: Bootstrap Design

- ▣ Phillips and Shi: the 95% quantile of the maximum of BSADF over τ_b consecutive dates sets the FWER at 5% for the bootstrap design: τ_b monitoring dates after a start-up of w_0 observations [Phillips & Shi \(2020\)](#)
- ▣ Residuals of $\Delta y_t = a + e_t$, drawn with replacement and multiplied by $N(0, 1)$ weights; paths of $w_0 + \tau_b - 1$ changes; 999 replications
- ▣ Here $\tau_b = 104$ weeks (two years); applied to BSADF over the full history, whose later dates search more start points, the 5% control is approximate, not proved
- ▣ The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

Appendix: Real Time vs Hindsight: Definitions

- Start: first exceedance of the chain of BSADF episodes that leads to the peak, dated in hindsight
- Confirmed: after L consecutive exceedances, the first date the episode is known in real time, but with w_0 and L set from the full-sample T (a truly real-time design fixes them ex ante)
- Leads in months (GameStop: months of calendar days); rise: from confirmation to the peak; fall: to the lowest close within 2 years (GameStop: 1 year)
- Bitcoin 2021: the explosive chain peaked at the local high of 13 Apr 2021 and ended in May 2021; the all-time high of 8 Nov 2021 (episodes table) came after a deep fall and a slower recovery that BSADF did not flag

Appendix: Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate; Astill et al. compare monitoring statistics with the maximum over a training period, so the false-positive rate is set for a chosen monitoring horizon, robust to time-varying volatility [Astill et al. \(2018\)](#)
- Bubbles and crashes: after a bubble is detected, monitor for the crash with the sign of the mean of Δy_t (explosive vs stationary regime); never signals a crash before it starts [Whitehouse, Harvey & Leybourne \(2023\)](#)
- Earlier detection: alternative variance standardisations keep the false-detection probability of existing procedures but find bubbles sooner (applied to Bitcoin and housing) [Whitehouse, Harvey & Leybourne \(2025\)](#)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery [Phillips & Shi \(2018\)](#)
- Alternatives to sup-ADF: a Chow-type break test from random walk to explosive has the highest power in Homm and Breitung [Homm & Breitung \(2012\)](#); recursive tests with a start-point search [Harvey, Leybourne & Sollis \(2015\)](#)

Appendix: LPPL in the Cryptocurrency and Romanian Markets

- ▣ Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash [Pele, Mazurencu & Nijkamp \(2013\)](#); crash modelling for the BVB [Pele & Mazurencu-Marinescu \(2012\)](#)
- ▣ Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles [Pele & Mazurencu-Marinescu-Pele \(2019\)](#), [Conda et al. \(2026\)](#)
- ▣ Dogecoin 2021: LPPL together with sentiment from social media [Pele \(2025\)](#)
- ▣ Common lesson: LPPL describes the acceleration well; the crash date remains the weak point

Appendix: Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood by prediction and update: $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1}) \varphi(r_t; \mu_j, \sigma_j^2)$, $\mathcal{F}_t$ = returns up to t , $\varphi(x; \mu, \sigma^2)$ the Normal density with mean μ and variance σ^2 [Hamilton \(1989\)](#)
- Kim smoother, backwards: $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \frac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$ [Kim \(1994\)](#)
- EM algorithm: with smoothed $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$ and $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$, the M-step is weighted least squares [Hamilton \(1990\)](#)
 - ▶ $\hat{\mu}_j = \frac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}$, $\hat{\sigma}_j^2 = \frac{\sum_t \xi_{tj} (r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}$, $\hat{p}_{ij} = \frac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- The likelihood is unbounded (one regime on a single return with $\sigma_j \rightarrow 0$) and has several local maxima (regime labels can switch): we take the best interior maximum from 20 random starts and check that no $\hat{\sigma}_j$ collapses towards 0, also in bootstrap refits
- Standard errors from the inverse numerical Hessian; they assume the number of regimes is known

Appendix: Why the LR Test for Regimes Is Non-Standard

- Under H_0 the regime means and variances coincide, so p_{11}, p_{22} are not identified (nuisance parameters present only under H_1); the information matrix is singular and some scores are identically zero [Hansen \(1992\)](#)
- Hansen: bounds on the standardised LR via empirical-process theory [Hansen \(1992\)](#)
- Carrasco, Hu and Ploberger: an optimal test needing only the estimate under H_0 , with bootstrap critical values [Carrasco, Hu & Ploberger \(2014\)](#)
- Same logic for two vs three regimes: the null model is then the fitted two-regime chain

Appendix: Case Study, Data and Method

- Data (Section 1.1): 51 580 408 arm's-length home sales from deeds records, 1995–2014, in 115 metropolitan areas (48% of the US housing stock); the cycle analysis covers 2000–2011, since realised holding periods need consecutive sales
 - ▶ Listings of homes for sale linked to the sales at property level in 57 of these areas
- Speculators (Section 1.2): short-term buyers resell within three years (realised holding period); non-occupant buyers give a mailing address different from the property address (102 areas)
- Lead-lag (Eq. 1, Figure 2): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$ for $k = -12, \dots, 48$ months
 - ▶ p : log price, demeaned by city; v : sales divided by the 2000 housing stock, demeaned by city and calendar month; implied correlation $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$, standard errors clustered by month
- Cross-section (Table 2): OLS across cities of the price boom (2000–2006), price bust (2006–2011), listings, volume and foreclosures on the speculative volume boom (2000–2005)
- Model (Sections 5–6): buyers extrapolate recent price growth and differ in horizon; sellers post list prices and not every listing sells (non-Walrasian market)
 - ▶ Counterfactuals: rational expectations, no speculators, transaction taxes

Appendix: Case Study, Complete Results

- Figure 2: the correlation between prices and lagged volume peaks at a lag of 24 months; Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005), short-term sales from 510 to 940 thousand, 39% of the rise (non-occupant purchases: 53%)
 - ▶ Interpretation: volume leads prices by about two years, and speculators account for much of the volume boom
- Table 2, Panel A: coefficient of the short-term sales volume boom in the price-boom regression 1.930 (standard error 0.297, $R^2 = 0.272$); one standard deviation more (12.9%, Table 1) means a price rise larger by 24.9 percentage points; price bust -0.571 (0.083), 7.4 points deeper
 - ▶ Interpretation: cities with more speculation had both larger booms and larger busts
- Table 2, Panels B–C (quantities in % of 2003 total volume): unsold listings in the quiet 5.961 (1.353), 76.9 points more per standard deviation; foreclosures 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas (Section 3.2)
 - ▶ Interpretation: consistent with speculators who bought in the boom listing in the quiet and failing to sell; that this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year goes with an 8.2 points larger short-term buyer share; Table 6: in the model, a 5% purchase tax on all buyers cuts the price bust from 8.2% to 1.1%, on non-occupants only to 5.8%
 - ▶ Interpretation: extrapolation draws short-horizon buyers in; short horizons, not non-occupancy, amplify the cycle

Appendix: Case Study, Lead–Lag Inference

- p and v demeaned over the full sample; implied correlation $\hat{\beta}_k \text{sd}(v_{t-k})/\text{sd}(p_t)$, $k = -12, \dots, 48$ months
- At $k = 0$ the correlation is 0.03, at $k = -12$ it is -0.34 : level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean: the “implied correlation” is a standardised slope, not a Pearson correlation, so bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series, doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory
- 1990–2011: the maximum moves to the edge of the grid, $k = 48$ (0.82)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design.”
 - ▶ Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing explanations, each with its own observable
 - ▶ “For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts.”
 - ▶ E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Implement GSADF with $r_0 = 0.01 + 1.8/\sqrt{T}$ on weekly Bitcoin closes from data/market and simulate its 95% critical value.”
 - ▶ Replicate first: GSADF = 3.85 and critical value 2.27 on 626 weeks, as in this chapter

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
 - ▶ “Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.”
 - ▶ Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?”
 - ▶ Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid p -values, all 12 cells, the alarms that failed
 - ▶ The prompts that shaped the design go into AI_USE.md

Appendix: What the Researcher Must Check

- **Full-sample information in date-stamping:** critical values, lag order or detrending fitted on the whole sample
 - ▶ Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail:** left-tail ADF critical values (about -2.86) used against an explosive alternative
 - ▶ Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility:** Monte Carlo critical values under constant variance distort the size
 - ▶ Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity:** many dates, assets and windows; some alarm is almost certain
 - ▶ Test: run the monitoring rule on simulated random walks and count the paths with at least one alarm
- **Overlapping outcomes:** 90-day windows overlap, so Fisher's exact test or a χ^2 test overstates significance; windows cut by the end of the data must be dropped
 - ▶ Test: block bootstrap or circular rotations of the alarm series, which keep the dependence of each series; rotations also need stationarity (mini-case)

Appendix: Mini-Case: What the Result Does and Does Not Say

- A large one-sided p is not evidence of **no** information: it only fails to support a **higher** crash probability after alarms
 - ▶ Both point estimates are **negative**: rotation p for a lower crash rate after alarms 0.07 (Bitcoin), 0.03 (Nasdaq 100); block-bootstrap CI and Newey–West test in Seminar 17, B6
 - ▶ A post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days, not an announced crash
- Assumptions behind the rotation p -values: under H_0 every circular shift of the alarm series (at least 26 steps, both ends joined) is as likely as the observed alignment
 - ▶ This is stronger than equal hit rates: it needs stationary series and alarms independent of the crash process; shared regimes or a time-varying crash rate can invalidate it
 - ▶ Hence the p -values are exploratory; their size should be checked by simulation under plausible null processes
- Outcomes: only dates whose whole horizon is observed (no censored windows at the end of the data)

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Cryptocurrency: Speculative Asset and Medium of Exchange
- ☐  Are Cryptos becoming alternative Assets?
- ☐  Statistics of Financial Markets

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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